These queries require a deep understanding of spatial, functional, and hierarchical relationships—information that is often invisible to conventional data stores. This is where the knowledge graph becomes essential. It acts as a semantic layer that stores and maps these relationships, from how components relate to specific spaces to how spaces are nested within buildings. This powerful semantic foundation allows our API to translate complex, human-intuitive questions into precise data retrieval operations across all of our distributed data stores.

The knowledge graph functions as an intelligent query router that first identifies relevant entities based on physical relationships, then orchestrates data retrieval from appropriate storage systems. When a user requests sensor information for a specific room, the query component follows this workflow shown in figure 5.11:

1. *Semantic resolution* - query the knowledge graph to identify the specified room, understanding that 'located within' may include direct placement or through spatial hierarchies.
2. *Entity discovery* - traverse relationships to find related entities—sensors, equipment, documents, or images, all connected to the same physical space.
3. *Data store routing*- use the discovered entity identifiers to query the appropriate time-series databases, document stores, or real-time data streams.
4. *Contextual assembly* - combine the retrieved data with the spatial and functional context from the knowledge graph to provide meaningful, structured responses.